

AUSTING (LIANGHAN) DONG

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EDUCATION

University of Waterloo — Bachelor of Computer Science (Co-op)

Sep 2022 – present (expected 2027)

- GPA: 3.7/4.0; President's Scholarship of Distinction; Faculty of Math Entrance Scholarship (\$25,000 CAD).

SKILLS

Languages: Python, C, C++, Java, R, SQL, JavaScript, TypeScript

AI / ML: PyTorch, TensorFlow, LangChain, vLLM, NumPy, Pandas, OpenCV

Frameworks & Cloud: React/Preact + TypeScript, Vue, Spring Boot, FastAPI, Flask, Redis, MySQL, AWS

EXPERIENCE

Undergraduate Research Fellow — University of Waterloo

May 2026 – Present

- Investigating how the gap between few-shot prompting and fine-tuning scales with LLM size and pretraining, benchmarking open model families (GPT-2, Qwen) across a multi-task benchmark.

Undergraduate Research Fellow — University of Waterloo

Sep 2025 – Dec 2025

- Fine-tuned vision-language models with attention regularization and human-labeled supervision; built attention visualization and labeling interfaces for multimodal transformers.

Undergraduate Research Fellow — University of Waterloo

Jan 2025 – Apr 2025

- Published a first-author paper in IEEE TVCG (IEEE VIS 2025) introducing AG-CAM, an attention-guided framework for interpreting vision-language model reasoning.
- Built ChartPADs, a multi-agent LLM pipeline (GPT-4o, Gemini 2.5 Pro) for persona-adaptive chart explanations; a 50-participant study showed +22% comprehension accuracy and +24% perceived clarity.

Data Analyst Intern — Microsoft

Jan 2024 – Apr 2024

- Designed classification pipelines for anomaly detection on Windows Defender and Azure error logs, improving early failure detection by over 15%; automated reporting pipelines for reliability metrics.

Software Developer — YSAIT

Sep 2023 – Dec 2023

- Built backend infrastructure for an AI-driven e-learning platform (Flask, GPT-3.5 API); developed an admin dashboard (Spring Boot, MyBatis, Redis, MySQL), cutting API latency by 30%; deployed on AWS EC2.

SELECTED PROJECTS & PUBLICATIONS

Scotiabank Fraud Detection Challenge — 1st place among 500+ participants

Jan 2023

- Led a team of 5 to 1st place with a predictive EDA + ML pipeline for financial risk analysis. [\[code\]](#)

Probing the Visualization Literacy of Vision Language Models

2025

- First-author IEEE TVCG paper; evaluated 6 open- and closed-source VLMs (ChartGemma, Janus, LLaVA, GPT-4o, Gemini). [\[paper\]](#) [\[code\]](#)

NASA Space Apps NLP Hackathon — Global Nomination

2022

- Built an NLP system for exploring scientific literature; globally nominated (Science Legacy track). [\[project\]](#)

AWARDS

- USA Computing Olympiad (USACO) — Gold Division, full marks (C++) 2021
- Euclid Math Contest — Top 1% 2022
- Statistics & Actuarial Science Competition Prize, University of Waterloo 2022